

VIRASATX

India's Heritage, Understood, Preserved & Experienced

- **Problem Statement ID** – SIH26197
- **Problem Statement**- Student Innovation—Ideas that showcase the rich cultural heritage and traditions of India.
- **Theme**- Heritage & Culture
- **PS Category**- Software
- **Organization**- AICTE
- **Project Pitch**- “An AI-powered digital heritage ecosystem that connects monuments, artifacts, traditions, places, people and knowledge into one discoverable cultural experience.”
- **Team ID**- 57385
- **Team Name**- Team VirasatX



❖ Proposed Solution:

• Fragmented Knowledge & Passive Discovery :

Heritage information is distributed across monuments, archives, museums and disconnected sources; digital heritage is often presented as isolated information.

• Low Discoverability & Context Gap :

Traditional crafts, artisans and living heritage are hard to discover; understanding wider connections to history, place and community is fragmented.

• Proposed Solution — Unified Discovery Platform :

VirasatX brings these discovery pathways together through one interconnected digital heritage experience.

• Heritage Collections & Artifact Exploration :

Detailed digital exploration of cultural artifacts, contextual narratives, and historical background.

• Chronological Cultural Timeline :

Chronological exploration of Indian heritage across major historical periods and epochs.

• Interactive Heritage Map :

Geographical exploration linking monuments, regional heritage sites, and cultural traditions.

• Living Traditions & Artisan Discovery :

Dedicated discovery pathway for traditional crafts, artisans, communities, and cultural practices.

• AI Heritage Guide & Visual Iconography AI :

AI-assisted heritage guidance for contextual exploration and AI-assisted visual iconography discovery.

• Manuscripts & Structured Learning :

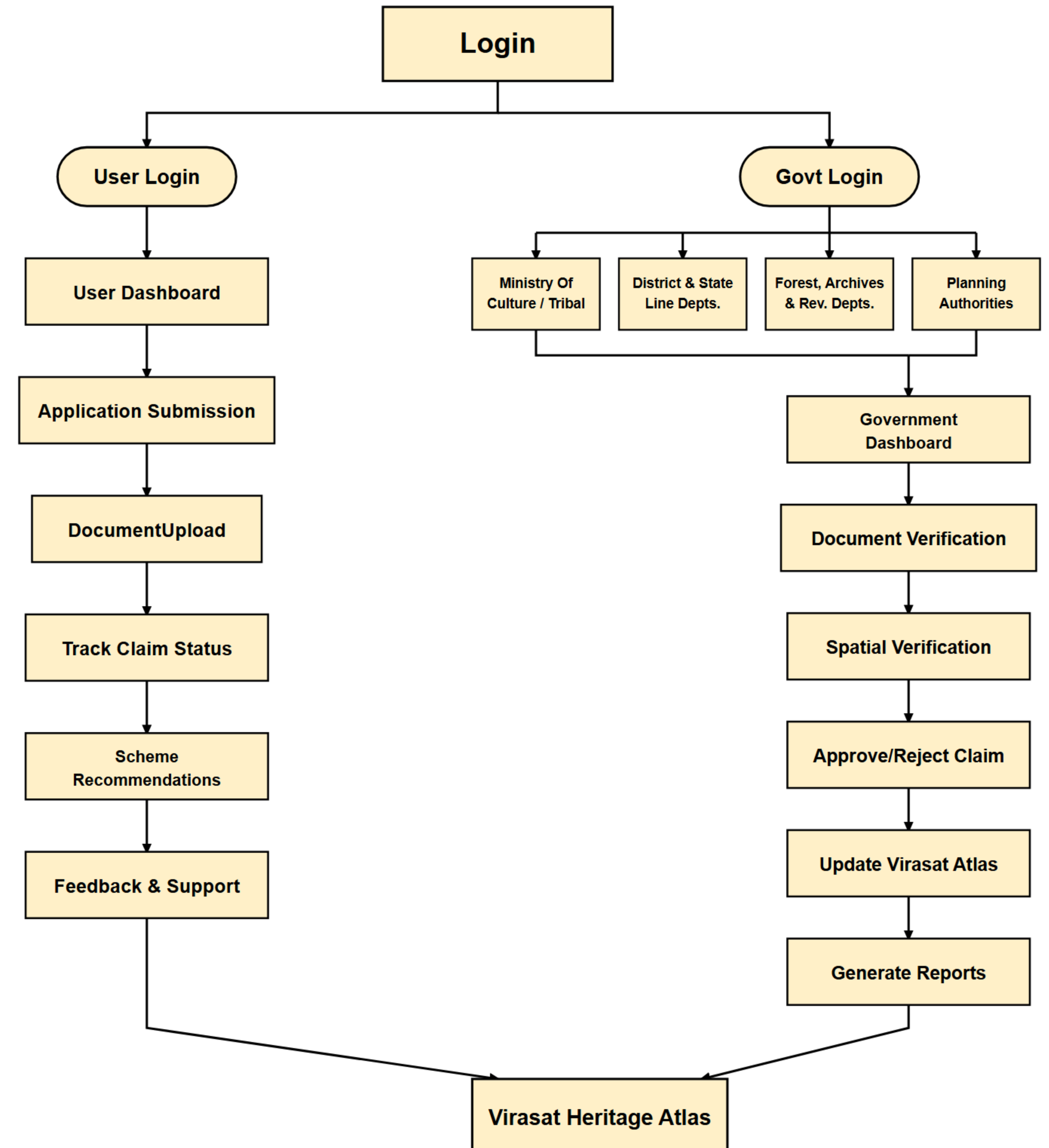
Dedicated exploration of textual and inscriptional heritage with structured learning and responsible discovery.

• Unique Value Proposition (UVP) :

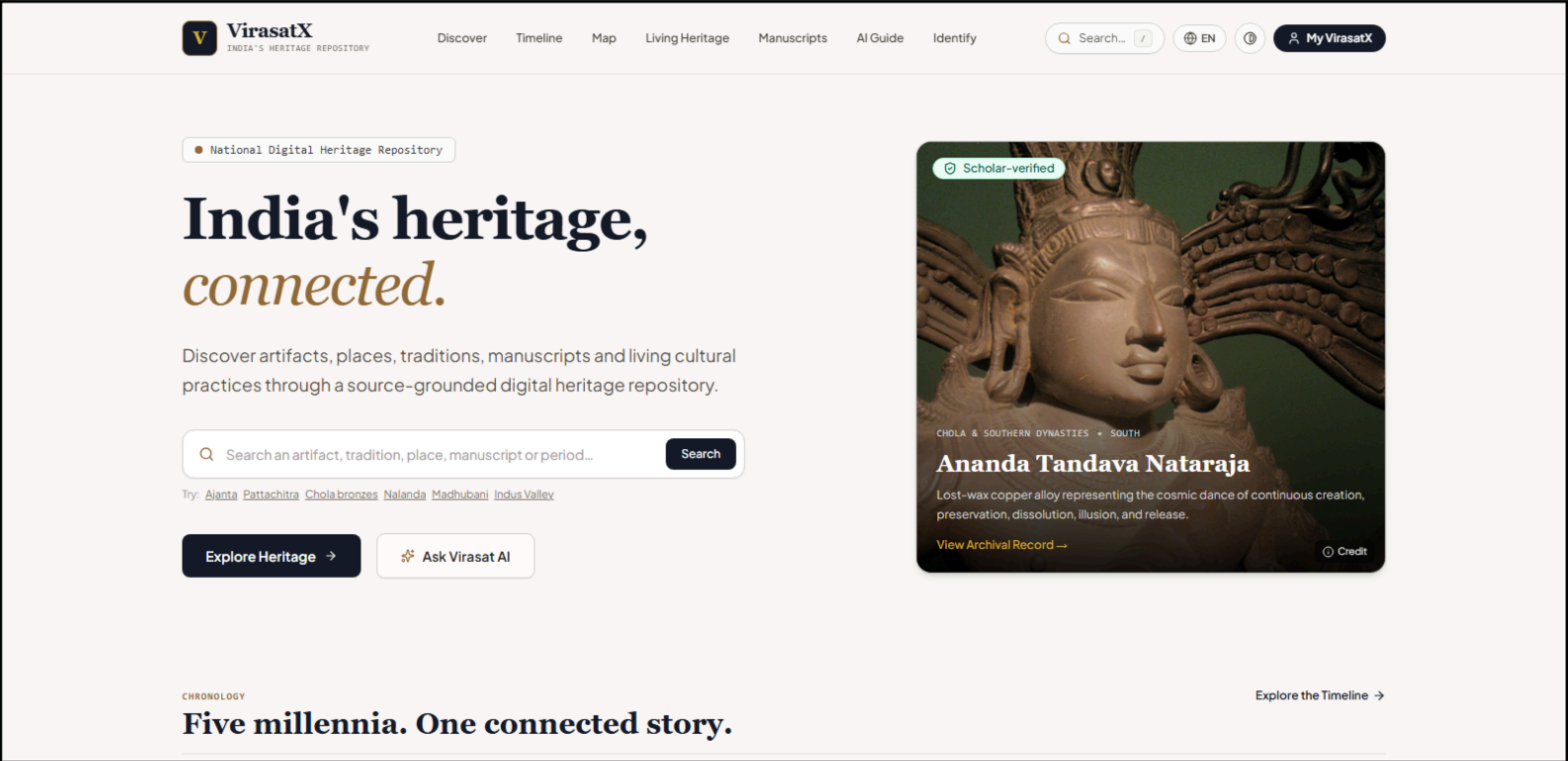
Artifact → Story → Period → Place → Tradition → Community → Learning → Responsible Experience.

• Cultural Context Focus :

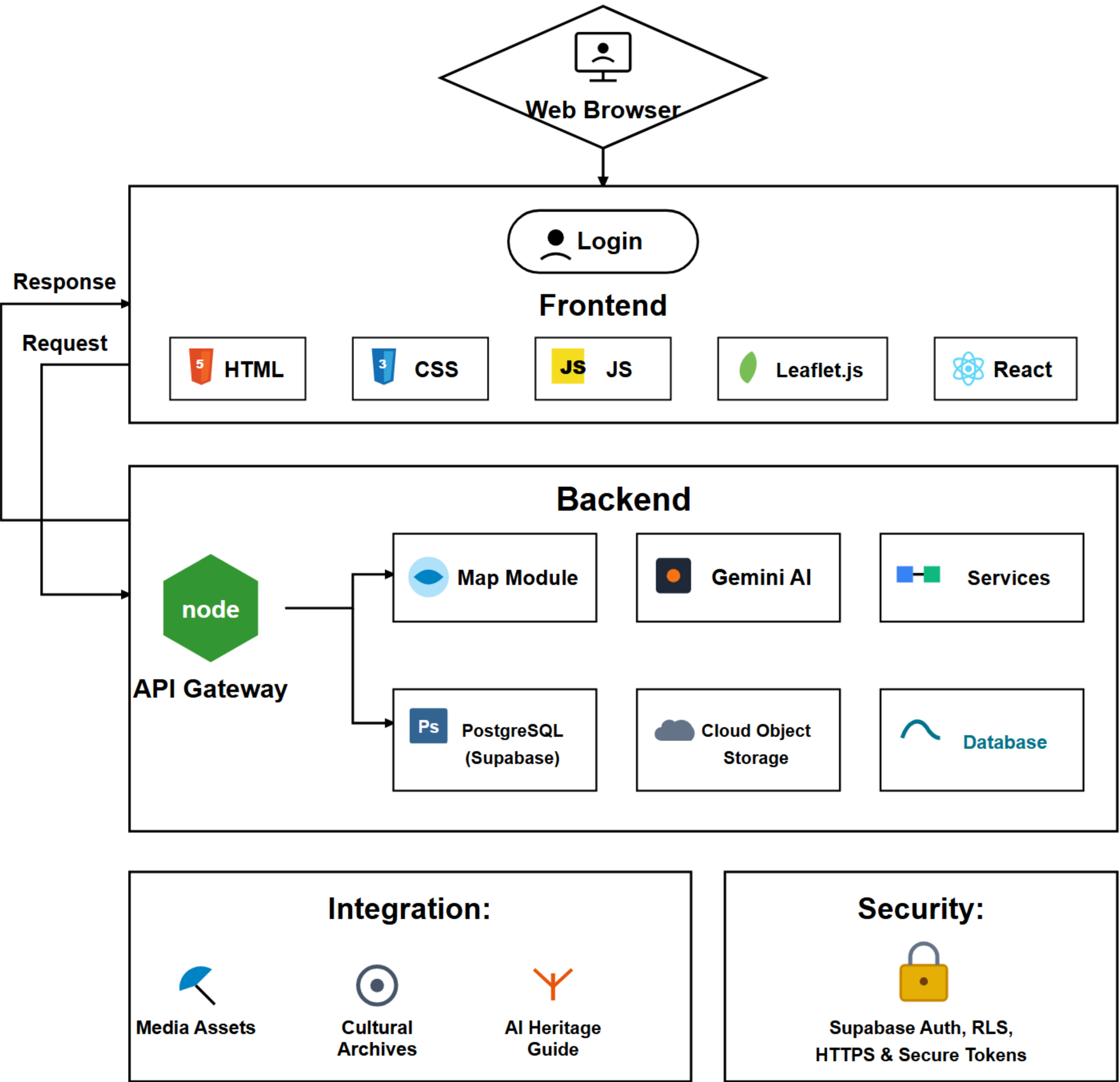
“VirasatX connects the cultural context around heritage — not just the heritage object itself.”



- **Frontend Framework:** React 19, TypeScript, Vite & Tailwind CSS with Motion.
- **AI Heritage Guide:** Google Gemini Multimodal AI for contextual cultural exploration.
- **Visual Iconography AI:** AI-assisted visual guidance for iconography and motif discovery.
- **Interactive Mapping:** Leaflet / OpenStreetMap for geographical heritage discovery.
- **3D Exploration Studio:** Three.js & WebGL for interactive 3D artifact exploration.
- **Voice Interaction:** Web Speech API for voice-assisted queries and audio guidance.
- **Backend & Database:** Supabase with PostgreSQL database and Row Level Security (RLS).
- **Secure Authentication:** Supabase Auth with role-based access control policies.
- **Cloud Deployment:** Vercel serverless edge deployment for global responsiveness.
- **Scalable Architecture:** Modular component hierarchy with structured content schemas.



User Interface: It is a draft version, subject to future updates.



❖ Core Capabilities:

• Virasat AI :

AI-assisted heritage guidance for contextual exploration and research-oriented discovery.

• Visual Iconography AI :

AI-assisted exploration of visual cultural and iconographic information.

• Heritage Atlas :

Interactive geographical discovery of heritage places and regions.

• Cultural Timeline :

Chronological exploration of Indian heritage across historical periods.

• Living Heritage :

Discovery pathway for traditional crafts, artisans, communities and cultural practices.

❖ Discovery Pathways & UVP:

• Manuscripts & Epigraphy :

Dedicated exploration of textual and inscriptional heritage.

• Artifact Exploration :

Detailed digital exploration of cultural artifacts and their context.

• Responsible Heritage :

Encourages informed and culturally responsible heritage discovery.

• Core Differentiator :

MOST PLATFORMS:

“What is this place?”

VIRASATX:

“How does this place connect to history, people, traditions, artifacts and living culture?”

• Context-First Experience :

“VirasatX connects the cultural context around heritage — not just the heritage object itself.”

❖ Feasibility & Impact:

1. Technical Feasibility:

- Modern React-based web architecture with Vite and Tailwind CSS.
- Modular AI capabilities powered by Google Gemini.
- Supabase-backed authentication and structured data services.
- Interactive mapping, 3D exploration, and multimedia.
- Responsive web interface deployed globally via Vercel.
- Expandable content architecture supporting new heritage categories.

2. Social & Cultural Impact:

• **Preservation & Discovery:**

Makes cultural information more discoverable through a unified digital experience.

• **Education:**

Transforms heritage discovery into an interactive learning experience.

• **Community Visibility:**

Creates a digital pathway for discovering living traditions, crafts and artisan communities.

• **Accessibility:**

Allows people to explore cultural knowledge beyond physical locations.

• **Responsible Discovery:**

Encourages understanding cultural context before visiting or interacting with heritage.

❖ Scalability & Future Scope:

1. Scalability Architecture:

• **Scalability Progression:**

VirasatX MVP → Expanded Heritage Records → More Regions & Languages → Institutional / Archive Integrations → Community Contribution → National-Scale Heritage Knowledge Platform.

- Modular architecture allows adding regions and categories without system redesign.
- Serverless edge hosting through Vercel and Supabase scales on demand.

2. Future Scope (Planned Enhancements):

- Indian-language expansion across additional regional languages.
- Larger institutional datasets from archives and museum records.
- Community contribution workflows for grassroots cultural discovery.
- Advanced visual recognition for iconography.
- Richer 3D / immersive cultural experiences.
- Expanded educational modules and curricula integration.
- Museum and archive integrations.

Note: The above items represent future scope and are not claimed as currently implemented.

1. Project & Prototype:

VirasatX — India's Heritage Repository

Live Prototype: <https://virasatxai.vercel.app/>

GitHub Repository: <https://github.com/ShrilCarpenter/VirasatX>

2. Research & Reference Sources:

Institutional sources consulted for taxonomy, cataloging schemas, and heritage documentation:

- Ministry of Culture (<https://www.indiaculture.gov.in/>)
- Archaeological Survey of India (ASI) (<https://asi.nic.in/>)
- National Archives of India (<https://nationalarchives.nic.in/>)
- National Mission on Monuments and Antiquities
- Museums of India (<https://museumsofindia.gov.in/>)
- Indira Gandhi National Centre for the Arts (IGNCA) (<https://ignca.gov.in/>)
- Vedic Heritage Portal (<https://vedicheritage.gov.in/>)
- UNESCO / Relevant Institutional Heritage Resources (<https://whc.unesco.org/>)

3. Closing Vision:

VIRASATX

“From scattered heritage knowledge to one connected cultural experience.”

Explore. Understand. Preserve.

Smart India Hackathon 2026 — Problem Statement ID: **SIH26197**